

Dominating Sets with Matching Outside *

Chih Yuan Lin¹, Jia Jie Liu², Yue Li Wang¹ and Chiun Chieh Hsu¹

¹Department of Information Management,

National Taiwan University of Science and Technology, Taipei, Taiwan

²Department of Information Management,

Shih Hsin University, Taipei, Taiwan

{ylwang}@cs.ntust.edu.tw

Abstract

In this paper, we introduce a new variant of domination called the outer-paired domination. For a graph $G = (V, E)$, an outer-paired dominating set $D \subseteq V$ is a dominating set of G such that the induced subgraph of $V \setminus D$ contains a perfect matching. The outer-paired domination number of a graph G is the cardinality of a minimum outer-paired dominating set of G . We show that finding the outer-paired domination number of a graph G is NP-hard on bipartite graphs. We also investigate the outer-paired domination problem for some special classes of graphs.

1 Introduction

Let $G = (V, E)$ be an undirected graph, where $V(G)$ and $E(G)$ are vertex and edge sets of G respectively. For simplicity, we also use V and E to represent $V(G)$ and $E(G)$, respectively, when the context is clear. The degree of a vertex v in a graph G is the number of edges incident with v and is denoted by $\deg(v)$. The minimum degree (respectively, maximum degree) of G is the minimum (respectively, maximum) degree among all vertices in G and is denoted by $\delta(G)$ (respectively, $\Delta(G)$). For a vertex $v \in V$, the open neighborhood of v is the set $N(v) = \{u \in V \mid uv \in E\}$.

The closed neighborhood of v is $N[v] = N(v) \cup \{v\}$. For a set $S \subseteq V$, let $N(S) = (\bigcup_{v \in S} N(v)) \setminus S$ and $N[S] = N(S) \cup S$.

A set D is a dominating set of a graph G if every vertex in $V \setminus D$ is adjacent to at least one vertex in D . The domination problem and its variants were extensively studied [2-7, 9-11, 13], e.g., independent domination [2], rainbow domination [3, 4], total domination [5], Roman domination [6], secure domination [7], k -domination [9,10], paired domination [12], weak Roman domination [13], and etc. We can find that all variants on the domination problems are adding some constraints on the dominating set itself before Cyman, in [8], investigated the outerconnected domination problem of a graph G which requires that the subgraph induced by $V \setminus D$ is connected. The outer-connected domination problem is subsequently studied in [1, 14, 15]. Inspired by [8], in this paper, we investigate a new variant of the domination problem called the outer-paired domination problem which is defined as follows.

Definition 1.1. An outer-paired dominating set (OPD-set for short) of a graph G is a dominating set D of G such that the induced subgraph of $V \setminus D$ contains a perfect matching. The outer-paired domination number, denoted by $\tilde{\gamma}_p(G)$, is the cardinality of a minimum OPD-set of G . If D is an OPD-set with $|D| = \tilde{\gamma}_p(G)$, then we say that D is a

* This work was supported in part by the National Science Council of Republic of China under contracts MOST 103-2221-E-128-002 and NSC 101-2221-E-011-038-MY3.

$\tilde{\gamma}_p$ -set. The outer-paired domination problem is to find out $\tilde{\gamma}_p(G)$ for a graph G .

The motivation of this new variant comes from the following practical resource allocation model. Consider a fault-tolerant Hadoop system which is modeled as a graph $G(V, E)$, where V represents the set of nodes and E is the set of links of node pairs. The nodes in $v \in V$ are classified into two roles: job-tracker and task-tracker. Every task-tracker is monitored by a job-tracker. When a task-tracker is failed, its job-tracker transfers its work to its backup task-tracker. Thus each task-tracker is adjacent to at least one job-tracker as well as a unique backup task-tracker. The objective is to minimize the cardinality of job-trackers over the entire network under the constraint that all task-trackers have a perfect matching.

The rest of this paper is organized as follows. In Section 2, we show that the outer-paired domination problem is NP-hard on bipartite graphs. In Section 3, we investigate the outerpaired domination number for some special classes of graphs, e.g., complete graphs, paths, cycles, and complete bipartite graphs. Finally, Section 4 contains our concluding remarks.

2 NP-hardness of outer-paired domination

In this section, we show that the outer-paired domination problem is NP-hard on bipartite graphs. We shall show that the vertex cover problem can be reduced to the outer-paired domination problem on bipartite graphs. A bipartite graph is a graph whose vertices can be partitioned into two independent sets U and V which are called the partite sets of G . Note that we use $G(U, V, E)$ to denote a bipartite graph where

U and V are the partite sets and E the edge set. A set S is a vertex cover of a graph G if each edge of G is incident to at least one vertex in S . The vertex cover problem is to find out a vertex cover with the least number of vertices.

Definition 2.1. Let $G = (V, E)$ with $V(G) = \{v_1, \dots, v_n\}$ and $E(G) = \{e_1, \dots, e_m\}$. Based on G , we construct a bipartite graph $H_G = (U, V, E)$ as follows. Let $U(H_G) = V_1 \cup V_3$, $V(H_G) = V_2 \cup V_4$, and $E(H_G) = E_1 \cup E_2 \cup E_3 \cup E_4$, where

$$V_1 = \{v_i^1 : v_i \in V(G), 1 \leq i \leq n\},$$

$$V_2 = \{v_j^2 : v_j \in V(G), 1 \leq j \leq n\},$$

$$V_3 = \{v_j^1 : e_j \in E(G), 1 \leq j \leq m\},$$

$$V_4 = \{v_i^2 : e_i \in E(G), 1 \leq i \leq m\},$$

$$E_1 = \{v_i^1 v_j^2 : v_i, v_j \in V(G), i = j \text{ or } v_i v_j \in E(G)\},$$

$$E_2 = \{e_i^1 e_j^2 : e_i \in E(G), 1 \leq i \leq m\},$$

$$E_3 = \{v_i^1 e_j^2 : v_i \in V(G), e_j \in E(G), e_j \text{ is incident to } v_i \text{ in } G \text{ for } 1 \leq i \leq n \text{ and } 1 \leq j \leq m\}, \text{ and}$$

$$E_4 = \{v_i^2 e_j^1 : e_j \in V(G), e_j \in E(G), e_j \text{ is incident to } v_i \text{ in } G \text{ for } 1 \leq i \leq n \text{ and } 1 \leq j \leq m\}.$$

Note that, in the construction of H_G , each edge $e_x = v_i v_j$ in $E(G)$ has a corresponding gadget in H_G as shown in Figure 1. Furthermore, if there is an edge $v_i^1 e_j^2$ in E_1 of H_G , then there is an edge $e_x = v_i v_j$ in $E(G)$.

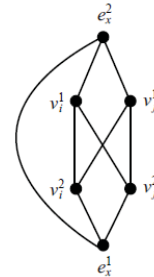


Figure 1: A gadget in H_G for an edge in $E(G)$.

Example 1. Figure 2 is an example to illustrate the construction of H_G from G . In Figure 2(a), there

is a graph G with 5 vertices and 6 edges. Figure 2(b) depicts the graph H_G in which $V_1 = \{v_1^1, \dots, v_5^1\}$, $V_2 = \{v_1^2, \dots, v_5^2\}$, $V_3 = \{e_1^1, \dots, e_6^1\}$ and $V_4 = \{e_1^2, \dots, e_6^2\}$. Note that there are 6 gadgets in Figure 2(b) since there are 6 edges in G .

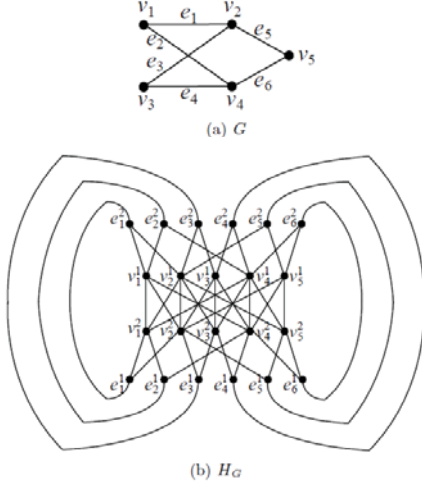


Figure 2: An example for illustrating graph H_G .

In the rest of this section, we use D to denote an OPD-set of graph H_G and $M(D)$ to denote a perfect matching of the subgraph induced by $(U(H_G) \cup V(H_G)) \setminus D$.

Lemma 2.2. *If $S = \{v_{a_i} : 1 \leq i \leq l\}$ is a vertex cover of G , then the set $D = \{v_{a_i}^1 : 1 \leq i \leq l\} \cup \{v_{a_i}^2 : 1 \leq i \leq l\}$ is an OPD-set of H_G .*

Proof. Since S is a vertex cover of G , by the construction of H_G , all vertices in $V_3 \cup V_4$ of H_G are dominated by D . Furthermore, all vertices in $(V_1 \cup V_2) \setminus D$ are also dominated by D . By the construction of E_1 and E_2 in H_G , there exist edges $v_i^1 v_i^2$ and $e_j^1 e_j^2$ for $1 \leq i \leq n$ and $1 \leq j \leq m$ in H_G . Thus a perfect matching can be found in the subgraph induced by $(U(H_G) \cup V(H_G)) \setminus D$. Therefore, the set D is an OPD-set of H_G .

Lemma 2.3. *Let D be an OPD-set of H_G . If $D \cup (V_3 \cup V_4) \neq \emptyset$, then there exists an OPD-set D'*

of H_G with $|D'| \leq |D|$ and $|D' \cap (V_3 \cup V_4)| < |D \cap (V_3 \cup V_4)|$.

Proof. Since $D \cap (V_3 \cup V_4) \neq \emptyset$, assume that there is an edge $e_\alpha^1 \in D$ for some $1 \leq \alpha \leq m$. Furthermore, let v_β and v_γ be the two endpoints of edge e_α in G . Thus e_α has a corresponding gadget (see Figure 3(a)). We distinguish the following two cases.

Case 1: $e_\alpha^2 \notin D$.

If $e_\alpha^2 \notin D$, then one of v_β^1 and v_γ^1 is matched with e_α^2 in $M(D)$. Assume without loss of generality that v_β^1 is matched with e_α^2 . We consider the following four subcases.

Case 1.1: $v_\gamma^1 \in D$ and $|\{v_\beta^2, v_\gamma^2\} \cap D| > 1$ (see Figure 3(b)).

In this case, set $D' = (D \setminus \{e_\alpha^1\}) \cup \{v_\beta^1\}$ and $M(D') = (M(D) \setminus \{e_\alpha^2 v_\beta^1\}) \cup \{e_\alpha^1 e_\alpha^2\}$ (see Figure 3(c)). It is easy to verify that D' is also an OPD-set of H_G with $|D'| = |D|$ and $|D' \cap (V_3 \cup V_4)| < |D \cap (V_3 \cup V_4)|$.

Case 1.2: $v_\gamma^1 \notin D$ and $|\{v_\beta^2, v_\gamma^2\} \cap D| \geq 1$ (see Figure 3(d)).

In this case, set $D' = (D \setminus \{e_\alpha^1\}) \cup \{v_\beta^1\}$ and $M(D') = (M(D) \setminus \{e_\alpha^2 v_\beta^1\}) \cup \{e_\alpha^1 e_\alpha^2\}$ (see Figure 3(e)). It is easy to verify that D' is also an OPD-set of H_G with $|D'| = |D|$ and $|D' \cap (V_3 \cup V_4)| < |D \cap (V_3 \cup V_4)|$.

Case 1.3: $v_\gamma^1 \in D$ and $|\{v_\beta^2, v_\gamma^2\} \cap D| = 0$.

Since $|\{v_\beta^2, v_\gamma^2\} \cap D| = 0$, there exists an edge $x v_\beta^2 \in M(D)$ with $x \in \{v_\omega^1, e_\omega^1\}$ (see Figures 3(f) and 3(g) for $x = v_\omega^1$ and $x = e_\omega^1$, respectively). We claim that e_ω^2 must be in D . Suppose to the contrary that e_ω^2 is not in D . Since e_ω^2 has only three neighbors, i.e., vertices v_β^1, v_ω^1 , and e_ω^1 in H_G , at least one of those neighbors must be in D to ensure that e_ω^2 is

dominated. We know that v_β^1 is matched with e_α^2 and one of v_ω^1 and e_ω^1 is matched with v_β^2 . Thus the vertex in $\{v_\omega^1, e_\omega^1\}$ which is not matched with v_β^2 must be in D . As a consequence, no vertex can be matched with e_ω^2 , a contradiction. Therefore, the claim holds. For the case where $x = v_\omega^1$, set $D' = (D \setminus \{e_\alpha^1 e_\omega^2\}) \cup \{v_\beta^1 v_\beta^2\}$ and $M(D') = (M(D) \setminus \{v_\beta^1 e_\alpha^2, v_\beta^2 v_\omega^1\}) \cup \{e_\alpha^1 e_\alpha^2, e_\omega^2 v_\omega^1\}$ (see Figure 3(h)). For the latter case, i.e, $x = e_\omega^1$, set $D' = (D \setminus \{e_\alpha^1 e_\omega^2\}) \cup \{v_\beta^1 v_\beta^2\}$ and $M(D') = (M(D) \setminus \{v_\beta^1 e_\alpha^2, v_\beta^2 e_\omega^1\}) \cup \{e_\alpha^1 e_\alpha^2, e_\omega^1 e_\omega^2\}$ (see Figure 3(i)). It is easy to verify that, in both cases above, D' is also an OPD-set of H_G with $|D'| = |D|$ and $|D' \cap (V_3 \cup V_4)| < |D \cap (V_3 \cup V_4)|$.

Case 1.4: $|\{v_\gamma^1, v_\beta^2, v_\gamma^2\} \cap D| = 0$.

In this case, let $xv_\beta^2 \in M(D)$ for $x \in \{v_\omega^1, e_\omega^1\}$. By using a similar argument as in Case 1.3, we can also get $D' = (D \setminus \{e_\alpha^1 v_\omega^2\}) \cup \{v_\beta^1 v_\beta^2\}$ and the similar $M(D')$'s as in Case 1.3 for $x = v_\omega^1$ and e_ω^1 . Thus this case holds.

Case 2: $e_\alpha^2 \in D$.

In this case, we claim that if there is a matched edge $v_i^1 v_j^2 \in M(D)$ with $v_i^1 \in \{v_\beta^1, v_\gamma^1\}$ and $v_j^2 \in \{v_\beta^2, v_\gamma^2\}$ (see Figure 3(j)), then the claim holds. To prove the claim, set $D' = (D \setminus \{e_\alpha^1 e_\alpha^2\}) \cup \{v_i^1 v_j^2\}$ and $M(D') = (M(D) \setminus \{v_i^1 v_j^2\}) \cup \{e_\alpha^1 e_\alpha^2\}$ (see Figure 3(k)). It is easy to verify that D' is also an OPD-set of H_G with $|D'| = |D|$ and $|D' \cap (V_3 \cup V_4)| < |D \cap (V_3 \cup V_4)|$. Thus the claim holds. By the claim, we assume that there is no matched edge in $M(D)$ whose both endpoints are in $\{v_\beta^1 v_\gamma^1, v_\beta^2 v_\gamma^2\}$. We consider the following subcases.

Case 2.1: $|\{v_\beta^1, v_\gamma^1\} \cap D| \geq 1$ and $|\{v_\beta^2, v_\gamma^2\} \cap D| \geq 1$.

In this case, let $D' = D \setminus \{e_\alpha^1 e_\alpha^2\}$ which is still a

dominating set of G . After that, set $M(D') = M(D) \setminus \{e_\alpha^1 e_\alpha^2\}$. It is clear that D' is also an OPD-set of H_G with $|D'| = |D|$ and $|D' \cap (V_3 \cup V_4)| < |D \cap (V_3 \cup V_4)|$ (see Figure 3(k)). Thus this case holds.

Case 2.2: $|\{v_\beta^1, v_\gamma^1, v_\beta^2, v_\gamma^2\} \cap D| = 1$.

Assume without loss of generality that $v_\beta^1 \in D$ (see Figure 3(l)). By the assumption that there is no matched edge in $M(D)$ whose both endpoints are in $\{v_\beta^1, v_\gamma^1, v_\beta^2, v_\gamma^2\}$, every vertex in $\{v_\gamma^1, v_\beta^2, v_\gamma^2\}$ is matched with some vertex not in $\{v_\gamma^1, v_\beta^2, v_\gamma^2\}$. Let $x \notin \{v_\gamma^1, v_\gamma^2\}$ be the matched vertex with v_β^2 . Now set $D' = (D \setminus \{e_\alpha^1 e_\alpha^2\}) \cup \{v_\beta^2, x\}$ and $M(D') = (M(D) \setminus \{xv_\beta^2\}) \cup \{e_\alpha^1 e_\alpha^2\}$ (see Figure 3(m)). It is clear that D' is also an OPD-set of H_G with $|D'| = |D|$ and $|D' \cap (V_3 \cup V_4)| < |D \cap (V_3 \cup V_4)|$.

Case 2.3: $|\{v_\beta^1, v_\gamma^1, v_\beta^2, v_\gamma^2\} \cap D| = 0$.

By the assumption that there is no matched edge in $M(D)$ whose both endpoints are in $\{v_\beta^1, v_\gamma^1, v_\beta^2, v_\gamma^2\}$ let xv_β^1 be a matched edge in $M(D)$ with $x \in \{v_\omega^2, e_\omega^2\}$. For the case where $x = e_\omega^2$, by using a similar argument as in Case 1.3, vertex e_ω^1 must be in D unless $e_\omega^1 v_\beta^2 \in M(D)$ (see Figure 3(n)). If $e_\omega^1 \in D$, then, by the reasoning in Case 1, this case holds. For the exception that $e_\omega^1 v_\beta^2 \in M(D)$, vertices v_ω^1 and v_ω^2 must be in D so that e_ω^1 and e_ω^2 are dominated. By setting $D' = (D \setminus \{e_\alpha^1 e_\alpha^2\}) \cup \{v_\beta^1 v_\beta^2\}$ and $M(D') = (M(D) \setminus \{v_\beta^1 e_\omega^2, v_\beta^2 v_\omega^1\}) \cup \{e_\alpha^1 e_\alpha^2, e_\omega^2 e_\omega^1\}$ (see Figure 3(o)), it is easy to verify that D' is also an OPD-set of H_G with $|D'| = |D|$ and $|D' \cap (V_3 \cup V_4)| < |D \cap (V_3 \cup V_4)|$. For the case where $x \in v_\omega^2$, if both e_ω^1 and e_ω^2 are in D , then, by using the same reasoning as in the claim at the beginning of Case 2, this case holds. It is clear that at least one of e_ω^1 and e_ω^2 must be in D ; otherwise, either e_ω^1 or e_ω^2 is not dominated. If only one of e_ω^1 and e_ω^2 in D , then, by Case 1, this case also holds. This concludes the proof

of the lemma.

□

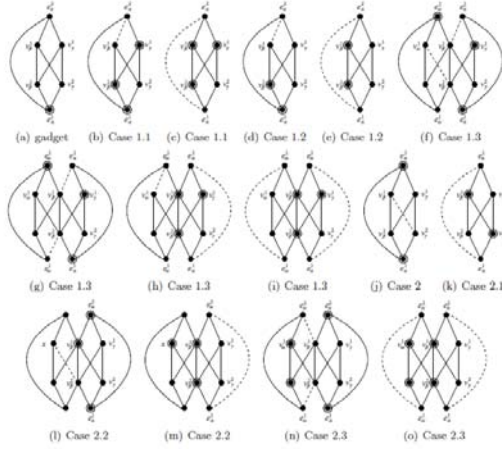


Figure 3: Illustrations for Lemma 2.3 in which a vertex in D is with a circle and a dotted line denotes a matched edge.

Theorem 2.4. *The outer-paired domination problem is NP-hard on bipartite graphs.*

Proof. We shall reduce the vertex cover problem on general graphs to the outer-paired domination problem on bipartite graphs. That is, there is a vertex cover of size k in G if and only if there is an OPD-set of size $2k$ in H_G . By Lemma 2.2, if $S = \{v_{a_i} : 1 \leq i \leq k\}$ is a vertex cover of G , then the set $D = \{v_{a_i}^1 : 1 \leq i \leq k\} \cup \{v_{a_i}^2 : 1 \leq i \leq k\}$ is an OPD-set of H_G . We can find that if $|S| = k$, then $|D| = 2k$.

For the converse, let D be an OPD-set of H_G . By Lemma 2.3, we may assume that $D \cap (V_3 \cup V_4) = \emptyset$. Let $D = D_1 \cup D_2$ with $D_1 \subseteq V_1$, $D_2 \subseteq V_2$, and $|D_1| \leq |D_2|$. Assume that $D_1 = \{v_{a_i}^1 : 1 \leq i \leq k\}$. Since there is no edge between sets D_2 and V_3 , all vertices in V_3 are dominated by D_1 . This implies that $D' = D_1 = \{v_{a_i}^2 : 1 \leq i \leq k\}$ is also an OPD-set of H_G . This further implies that the set $S = \{v_{a_i} : 1 \leq i \leq k\}$ is a vertex cover of G . Note that if $|D'| = 2k$, then $|S| = |D_1| = |D'| / 2 = k$. This completes the proof.

3 Outer-paired domination in some special graphs

In this section, we investigate the outer-paired domination number for some special classes of graphs including complete graphs, cycles, paths, and trees.

Let K_n , C_n , and P_n denote a complete graph, a cycle, and a path of order n , respectively. Let $K_{m,n}$ denote a complete bipartite graph $G(U, V, E)$ with $|U| = m$, $|V| = n$, and $n \geq m$. The outer-paired domination numbers of K_n , C_n , P_n , and $K_{m,n}$ are described in Propositions 3.1-3.4.

Proposition 3.1.

$$\tilde{\gamma}_p(K_n) = \begin{cases} 1, & \text{if } n \text{ is odd} \\ 2, & \text{if } n \text{ is even.} \end{cases}$$

Proposition 3.2.

$$\tilde{\gamma}_p(C_n) = \lfloor \frac{n}{3} \rfloor + (n \pmod{3}).$$

Proposition 3.3.

$$\tilde{\gamma}_p(P_n) = \begin{cases} \lfloor \frac{n}{3} \rfloor + 2, & \text{if } n \equiv 0, 2 \pmod{3} \\ \lfloor \frac{n}{3} \rfloor + 1, & \text{if } n \equiv 1 \pmod{3}. \end{cases}$$

Proposition 3.4.

$$\tilde{\gamma}_p(K_{m,n}) = n - m + 2.$$

4 Conclusion

In this paper, we propose a new variant of the domination problem, called the outer-paired domination problem, on graphs. We show that the outer-paired domination problem is NP-hard on bipartite graphs. As a future study, it is interesting to investigate the relation between the outer-paired domination number and other types domination numbers.

References

- [1] M.H. Akhbari, R. Hasni, O. Favaron, H.

- Karami, and S.M. Sheikholeslami, On the outer-connected domination in graphs, *Journal of Combinatorial Optimization* 26 (2013) 10-18.
- [2] R.B. Allan and R.C. Laskar, On domination and independent domination numbers of a graph, *Discrete Mathematics* 23 (1978) 73-76.
- [3] B. Brešar, M.A. Henning and D.F. Rall, Rainbow domination in graphs, *Taiwanese Journal of Mathematics* 12 (2008) 213-225.
- [4] B. Brešar and T.K. Šumenjak, On the 2-rainbow domination in graphs, *Discrete Applied Mathematics* 155 (2007) 2394-2400.
- [5] E.J. Cockayne, R.M. Dawes, and S.T. Hedetniemi, Total domination in graphs, *Networks* 10 (1980) 211-219.
- [6] E.J. Cockayne, P.M. Dreyer Jr., S.M. Hedetniemi, and S.T. Hedetniemi, On Roman domination in graphs, *Discrete Mathematics* 278 (2004) 11-22.
- [7] E.J. Cockayne, P.J.P. Grobler, W. Grundlingh, J. Munganga, and J.H. van Vuuren, Protection of a graph, *Utility Mathematics* 67 (2005) 19-32.
- [8] J. Cyman, The outer-connected domination number of a graph, *The Australasian Journal of Combinatorics* 38 (2007) 35-46.
- [9] J.F. Fink and M.S. Jacobson, n-domination in graphs, *Graph Theory with Applications to Algorithms and Computer Science*, John Wiley and Sons, New York (1985), 282-300.
- [10] J.F. Fink and M.S. Jacobson, On n-domination, n-dependence and forbidden subgraphs, *Graph Theory with Applications to Algorithms and Computer Science*, John Wiley and Sons, New York (1985), 301-311.
- [11] T.W. Haynes, S.T. Hedetniemi, P.J. Slater, *Fundamentals of Domination in Graphs*, Marcel Dekker, New York, 1998.
- [12] T.W. Haynes, P.J. Slater, Paired domination in graphs, *Networks* 32 (1998) 199-206.
- [13] M.A. Henning and S.T. Hedetniemi, Defending the Roman Empire - A new strategy, *Discrete Mathematics* 266 (2003) 239-251.
- [14] H. Jiang and E. Shan, Outer-connected domination in graphs, *Utilitas Mathematica* 81(2010) 265-274.
- [15] J.M. Keil and D. Pradhan, Computing a minimum outer-connected dominating set for the class of chordal graphs, *Information Processing Letters* 113 (2013) 552-561.